Who Benefits Most and Why? Evaluating HUD Grant Effectiveness on Childhood Lead Exposure with Causal Machine Learning

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Abstract

This study evaluates the effectiveness of HUD-administered lead hazard control grants on reducing childhood blood lead levels (BLLs) in Washington State. Using a county-year panel dataset from 2000 to 2023, we apply Causal Forest to estimate heterogeneous treatment effects (CATEs) across 39 counties. On average, grants reduced BLL exceedance by 1.42 percentage points. However, effects varied widely—counties with older housing and lower income saw greater reductions, while high-poverty areas experienced diminishing returns. Fixed-effects OLS and GAM models confirm that housing age, income, and unemployment mediate policy impact. Our results highlight the importance of targeting federal resources where structural vulnerability and implementation capacity align. Causal machine learning provides new insights into where HUD grants work best, offering guidance for more equitable and effective environmental health interventions.

Keywords: *HUD lead reduction grant, Children Lead Exposure, Machine Learning*

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I. Introduction

Childhood exposure to lead remains one of the most pressing and persistent public health threats in the United States. According to the Centers for Disease Control and Prevention (CDC), approximately 2.5% of U.S. children aged 1–5 have blood lead levels at or above the current reference value of 3.5 µg/dL. The CDC emphasizes that no safe blood lead level in children has been identified, as even minimal exposure is associated with developmental delays, cognitive impairment, and behavioral issues (CDC 2021).[3] Despite substantial nationwide reductions in lead exposure following the removal of lead from gasoline and household paint, millions of U.S. homes—particularly those built before 1978—continue to pose risks to children's health (Aizer et al., 2018; Reyes, 2015b; Grandjean et al., 2014).

To mitigate these risks, the U.S. Department of Housing and Urban Development (HUD) has implemented various lead hazard control programs. These efforts include regulatory bans in the 1970s and over $2 billion in grant-based funding to support lead abatement in low-income residential units. While these initiatives have made measurable progress, existing evaluations often rely on average treatment effects (ATE), obscuring whether grant effectiveness varies across communities. In particular, it remains unclear whether HUD grants reduce lead exposure uniformly—or whether certain counties benefit more due to underlying socioeconomic and structural conditions.

Understanding this heterogeneity is critical for designing equitable and cost-effective interventions. If federal lead hazard grants disproportionately benefit counties with specific characteristics—such as higher poverty rates, older housing stock, or denser populations—then resource allocation strategies must be adapted accordingly. Recent advances in causal machine learning provide the tools to explore these distributional questions in greater depth.

This study applies the Causal Forest algorithm to estimate the Conditional Average Treatment Effects (CATEs) of HUD grants on childhood blood lead levels (BLLs) across counties in Washington State. We construct a county-year panel from 2000 to 2023, integrating BLL surveillance data, housing characteristics, and socioeconomic indicators. Unlike conventional parametric models, Causal Forest enables flexible, data-driven estimation of heterogeneous treatment effects without imposing restrictive functional forms. We complement this with linear and non-linear regression models to assess variable-level associations with both CATE and BLL outcomes.

These methods enable us to answer a central policy question: **Where and for whom do HUD grants work best?** We identify key moderating variables—including household income, poverty rates, housing age, population density, and demographic characteristics—and provide empirical guidance on how to better target limited federal resources. By comparing results from causal machine learning to traditional regression methods, we demonstrate the added value of modern inference techniques in complex policy environments.

[3] https://www.cdc.gov

This paper contributes to the growing literature on data-driven environmental policy and health equity. By uncovering spatial heterogeneity in policy effectiveness, we provide both methodological insights and actionable guidance for improving lead exposure prevention efforts.

The remainder of this paper is organized as follows. Section II reviews the existing literature on childhood lead exposure and HUD's lead hazard mitigation programs. Section III describes our study area, key policy background, and data context. Section IV presents the data sources and summary statistics. Section V outlines our empirical strategy, including both machine learning and econometric models. Section VI reports the results. Section VII discusses limitations and directions for future research. Section VIII concludes.

II. Literature Review

The detrimental impact of lead exposure on children's health and cognitive development has long been a subject of public health research. Extensive literature documents the irreversible consequences of elevated blood lead levels (BLLs), including diminished IQ, learning disabilities, and behavioral disorders (Lanphear et al., 2002; Canfield et al., 2003; Grandjean et al., 2014). Recognizing the persistence of lead paint hazards, particularly in older housing stock, the U.S. Department of Housing and Urban Development (HUD) initiated various grant-based programs aimed at mitigating these risks through lead hazard control interventions.

Several large-scale studies evaluate the effectiveness of these HUD-supported programs. Clark et al. (2011) examine the impact of HUD-supported lead hazard interventions on children's blood lead levels (BLL) using longitudinal data from 12 U.S. sites. They report BLL reductions of up to 30% within one-year post-intervention, with sustained effects over three years. Interventions targeting **exterior paint** were notably more effective than interior-only efforts. The study highlights the importance of **primary prevention** and shows meaningful benefits even among children with low initial BLLs, providing a strong empirical foundation for evaluating HUD program effectiveness.

Similarly, Strauss et al. (2005) analyzed blood-lead surveillance data from four Massachusetts communities to assess the impact of HUD-funded lead hazard control (LHC) treatments. Using matched treated and untreated housing units and over 1,300 children, they found that BLLs in HUD-treated units declined by up to 50%, nearly twice the rate observed in untreated controls. The study used robust matching and longitudinal models, demonstrating the effectiveness of LHC programs even after adjusting for background trends in lead exposure.

Complementing these findings, Gazze (2016) employs a difference-in-differences framework to show that state-level lead abatement mandates reduced the incidence of elevated blood lead levels (EBLLs) by 29%, and lowered special education enrollment by 8.1%, suggesting substantial long-term educational and fiscal benefits. However, she also highlights concerns about uneven enforcement and potential displacement effects, as families with young children may be pushed out of older housing stock.

Local policies also show promise. Korfmacher et al. (2013) evaluated the implementation and impact of Rochester, New York's local lead law, which mandated lead inspections for pre-1978 rental housing. Using inspection records, blood lead level (BLL) surveillance data, and landlord surveys, the study found that BLL prevalence among children declined from 8.3% to 4.4% in the two years following the law's implementation. Notably, 94% of inspected units passed visual inspection and 89% passed dust-wipe tests, suggesting widespread compliance. The law had no significant negative effect on the rental housing market, and the authors emphasize the importance of sustained enforcement, periodic inspections, and interagency collaboration to maintain long-term effectiveness.

A growing literature has linked early childhood lead exposure to persistent deficits in human capital accumulation. Sorensen et al. (2019) exploit variation in HUD lead hazard control grant timing and funding intensity across counties to show that the program significantly reduced elevated blood lead levels and improved elementary school test scores, particularly for Hispanic children. Using matched individual-level data from Rhode Island, Aizer et al. (2018) demonstrate that even modest reductions in blood lead levels during early childhood lower the probability of scoring substantially below proficiency in reading and math by nearly one percentage point per µg/dL. Similarly, Reyes (2015) leverages cohort-level variation in lead exposure across Massachusetts school districts to show that post-policy declines in childhood lead levels account for a 1–2 percentage point improvement in standardized test scores. Collectively, these studies highlight the role of environmental health shocks in explaining educational inequality and provide empirical support for integrated housing-health interventions aimed at disadvantaged populations.

Recent advancements in machine learning (ML) have enabled more accurate risk prediction of lead exposure. Recent econometric studies have increasingly highlighted the value of machine learning methods for causal inference, especially when estimating average and heterogeneous treatment effects in high-dimensional settings. While previous literatures studied the effectiveness of HUD's grant program with various traditional econometrical models, we did not find any literatures directly employed machine learning model to measure the efficiency of the grant program. Mulhern et al. (2023) applied Bayesian network models to identify high-risk childcare facilities for water lead contamination, outperforming traditional rule-based heuristics by up to 213% in precision. These models accounted for complex interactions between facility characteristics, water source, and socioeconomic variables. They emphasized the importance of interpretable, data-driven lead risk prediction tools that public health agencies can operationalize at scale. These efforts signal a shift from generalized screening methods to targeted, cost-effective interventions guided by ML-informed models.

While prediction-based ML models have demonstrated clear value, they do not answer causal questions—namely, what would have happened *in the absence* of a policy intervention. To bridge this gap, the field of **Causal Machine Learning (CML)** has emerged, combining ML's flexibility with the rigor of causal inference. Athey and Imbens (2019), and later Lechner (2023), formalized frameworks such as Causal Forest and Double Machine Learning (DML) to estimate heterogeneous treatment effects (HTEs) without relying on restrictive parametric assumptions. Baiardi and Naghi (2024) showed that CML methods offer superior robustness in high-dimensional settings, outperforming traditional regression models when estimating both ATE and CATE. They re-examine influential empirical studies using causal machine learning techniques—primarily Double Machine Learning (DML) and Causal Forests—to demonstrate their advantages over conventional approaches. Their findings show that causal ML methods not only yield more robust treatment effect estimates under nonlinearity and high-dimensional confounding but also uncover previously undetected heterogeneity in treatment responses. By comparing results from original OLS-based analyses to DML and causal forest implementations, the authors reveal that ML-based estimates often have greater precision and expose interaction effects that traditional models overlook. The paper offers strong empirical support for the integration of modern causal ML tools into applied policy evaluation, particularly when treatment effect heterogeneity is of central interest. Tiffin (2019) further demonstrated the interpretability benefits of Causal Forest models in economic policy applications, including identifying country-specific vulnerability to financial crises.

In the context of lead exposure policy, these tools allow researchers to assess both the average treatment effect and identify which subpopulations benefit most. Building on this emerging literature, the present study applies Causal Forest to estimate heterogeneous effects of HUD lead hazard grants across counties in Washington State. By integrating demographic, socioeconomic, and housing characteristics into a panel structure, this analysis contributes to bridging the gap between traditional program evaluation and machine-learning-assisted policy optimization.

III. BACKGROUND INFORMATION

Study Area: Counties in Washington State

This study focuses on the 39 counties within the state of Washington, a geographically and socioeconomically diverse region in the Pacific Northwest. The state exhibits considerable heterogeneity in housing stock, population density, and income distribution. Urban centers such as King and Pierce Counties are densely populated with a mix of older and newer housing, while rural counties like Garfield and Columbia have older housing infrastructures and limited access to remediation resources. These structural disparities provide a compelling setting for analyzing the heterogeneous effectiveness of environmental health policies across space.

Historically, Washington has faced elevated risks of childhood lead exposure due to its large stock of pre-1980 homes and reliance on well water systems in certain rural communities. According to the Washington State Department of Health, multiple counties continue to report blood lead levels (BLL) above the CDC threshold, particularly in areas with concentrated poverty and aging infrastructure. These patterns make Washington State an ideal case for investigating where and for whom federal lead remediation policies are most effective.

In recognition of these risks, the U.S. Department of Housing and Urban Development (HUD) has designated several Washington counties and cities—including **King County, Seattle, Tacoma, Spokane, Snohomish County, Whatcom County,** and **Yakima County**—as among the nation's **"Highest Lead-Based Paint Abatement Needs Areas"**, based on the prevalence of pre-1940 occupied rental housing units (HUD, 2021).[4] This designation reflects both the age of the housing stock and the vulnerability of resident populations, particularly children in low-income households. These jurisdictions were prioritized for federal lead hazard reduction funding due to their elevated risk of childhood lead exposure and structural barriers to abatement.

[4] https://www.hud.gov/sites/dfiles/HH/documents/LHRGrant_FY2021_NOFO_Appendices_4921.pdf

The inclusion of these counties in HUD’s national high-priority list not only underscores the **public health significance of lead exposure in Washington State** but also highlights the potential for spatially targeted interventions to yield meaningful reductions in blood lead levels (BLL). It further supports the rationale for analyzing heterogeneous treatment effects (HTEs) across counties, as both the **baseline housing risk** and the **allocation of federal resources** differ markedly within the state.

Furthermore, according to the Washington State Department of Health, only 7% of children under the age of six were tested for lead in Washington.[5] This testing rate is significantly lower than the national average.[6] The low testing rate suggests that the actual extent of lead exposure may be underestimated and that the issue does not receive sufficient public attention. This under-detection makes a study of this kind even more valuable by helping to illuminate the true distribution and impact of lead exposure across communities.

Figure 1 and Figure 2 in **Section IV** (Data) display the county-level distribution of children testing above the BLL threshold for two time periods: 2000–2004 and 2019–2023. These maps illustrate shifts in exposure patterns and regional disparities over time.

Table 1 presents the top five counties in Washington ranked by four key characteristics: population size, proportion of older homes (built before 1989), median household income, and the rate of children with BLLs exceeding the CDC threshold. These indicators provide an overview of the spatial and structural variation that motivates our heterogeneous treatment effects analysis. A detailed map of all 39 counties in Washington is provided in the APPENDIX A to visually support our spatial analysis.

TABLE 1. Top 5 County Profiles for Each Factor

	Population	Rate of old houses	Median income	Rate of children BLL exceeding the threshold
1	King	Franklin	King	Asotin
2	Pierce	Clark	Snohomish	Okanogan
3	Snohomish	Grant	Kitsap	Spokane
4	Spokane	Lewis	Pierce	Stevens
5	Clark	Wahkiakum	Clark	King

Policy Overview: HUD Lead Hazard Reduction Grant Programs

The U.S. Department of Housing and Urban Development (HUD) has administered several grant programs aimed at reducing lead-based paint hazards in residential housing, especially for low-income families with young children. Since the early 1990s, HUD's **Lead Hazard Control Grant Program (LHCGP)** and the **Lead Hazard Reduction Demonstration Grant Program (LHRDGP)** have awarded over $2 billion nationwide for inspection, risk assessment, interim controls, and abatement activities. These programs target homes built before 1978—the year when the federal government banned the residential use of lead-based paint.

[5] https://doh.wa.gov/data-and-statistical-reports/washington-tracking-network-wtn/lead-risk-and-exposure?utm_source=chatgpt.com
[6] https://mdinspectionpros.com/lead-poisoning-statistics-by-state/

Grants are competitively awarded to state and local governments based on factors such as housing risk profiles, program capacity, and proposed remediation strategies. In some years, funding is distributed to state-level agencies rather than individual counties, which may introduce ambiguity in intra-state allocation. Washington State has received HUD funding in multiple grant cycles, with some counties benefiting more consistently than others. However, little is known about the differential impact of these grants across communities, particularly when accounting for differences in poverty, housing age, and local implementation capacity.

Despite these substantial investments, relatively little is known about how the effectiveness of these grant programs varies across communities with differing socioeconomic and structural characteristics. In particular, questions remain as to whether grant impacts are uniformly distributed or whether certain counties experience stronger benefits due to underlying conditions such as poverty prevalence, housing age, or implementation infrastructure.

To address these questions, this study links the institutional background of HUD's lead remediation efforts with high-resolution county-year panel data. Through this integration, we examine the heterogeneous impacts of lead mitigation grants across Washington's counties.

Table 2 in the following DATA section summarizes the major lead hazard control grants received by Washington counties since 1997, including program type, award year, and amount. This institutional context provides a foundation for evaluating variation in policy intensity and implementation over space and time.

IV. DATA

To estimate the heterogeneous effects of HUD lead hazard control grants, we construct a county-year panel dataset for Washington State covering the period from 2000 to 2023. The dataset integrates multiple sources of demographic, socioeconomic, environmental, and policy information.

Blood Lead Level (BLL) Data

Data on childhood blood lead levels (BLLs) were obtained from the Washington State Department of Health (DOH), which publishes county-level surveillance statistics. The dataset includes the number of children tested, the number of cases exceeding the 5 μg/dL threshold, the total population of children under 72 months, and the percentage of tested children exceeding the threshold, spanning the years 2000 to 2023.

These data are reported in one-year, three-year, and five-year intervals. For analytical consistency and to improve measurement reliability—particularly in counties with small populations—we utilize the five-year aggregated data. To temporally align lead exposure data with policy implementation periods, we assign each multi-year observation to the midpoint year of the reporting interval (e.g., data from 2000–2004 is coded as 2002). This approach reflects the delayed nature of policy effects and mitigates annual volatility in the exposure data.

Figures 1 and 2 present the spatial distribution of BLL exceedance rates across Washington counties for two selected periods 2000 – 2004 and 2019 – 2023. Darker shading indicates a higher proportion of children exceeding the BLL threshold, highlighting regional disparities in lead exposure risk.

***Figure 1**. Percentage of Blood Lead Tests with Results ≥5µg/dL Per Year for Children <72 Months of Age (2000-2004)*

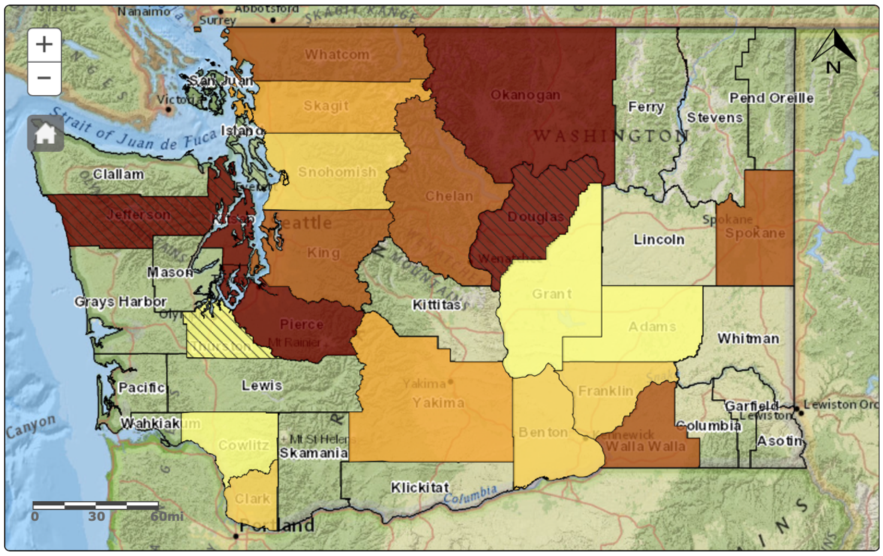
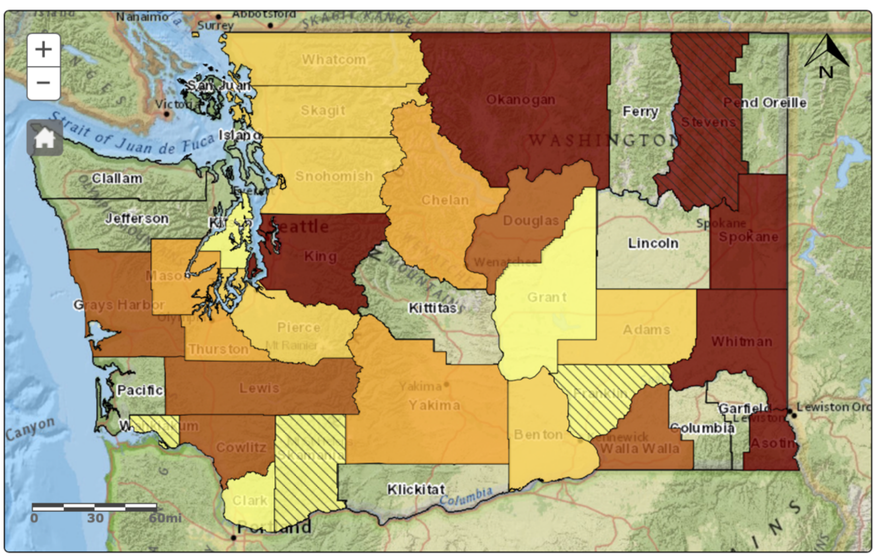


***Figure 2**. Percentage of Blood Lead Tests with Results ≥5µg/dL Per Year for Children <72 Months of Age (2019-2023)*



HUD Lead Grant Data

Historical records of lead hazard reduction grants were collected from the U.S. Department of Housing and Urban Development (HUD) program archive, which provides annual information on grantee jurisdictions, program types, and awarded amounts dating back to 1997. We aggregate the total grant amounts received by each county for each year.

In cases where funding was awarded to multi-county collaborations or non-county-specific entities (e.g., housing authorities or the state government), we allocate the grant amount proportionally based on the geographic scope of the service area, as documented by HUD. In particular, for the years 2005, 2012, and 2018, grants were awarded to the State of Washington without any disaggregated information regarding county-level allocation. For these years, we distribute the funds across counties in proportion to their population size.

Table 2 below summarizes major HUD lead hazard control grant allocations to Washington jurisdictions across selected years. These allocations represent a mix of direct municipal funding and broader state-level appropriations. They form the basis for constructing the policy treatment variable in our empirical analysis.

TABLE 2. HUD Lead Grant List

Year	Location	Name of the program	Amount
2000	City of Bellingham (Opportunity Council)	Lead-Based Paint Funding	$354,192
2001	Seattle/King County Public Health Department	Lead Hazard Control and Healthy Homes	$937,879
2004	City of Spokane	Lead Hazard and Healthy Homes grants	$2,290,954
2005	State of Washington	Lead Based Paint Hazard Control Grant Program	$3,000,000
2012	Washington State Department of Commerce	Lead Based Paint Hazard Control Grant Program	$2,480,000
2012	City of Spokane	Lead Based Paint Hazard Reduction Demonstration Grant Program	$2,400,000
2016	Yakima County	Lead Based Paint Hazard Control	$1,237,500
2018	State of Washington	Lead Based Paint Hazard Reduction (LBPHR) Amount	$121,741,969
2018	State of Washington	Healthy Homes Supplemental Funding (HHI) Amount	$17,620,400
2021	Housing Authority of the City of Seattle	PUBLIC HOUSING *Funding*	$567,256

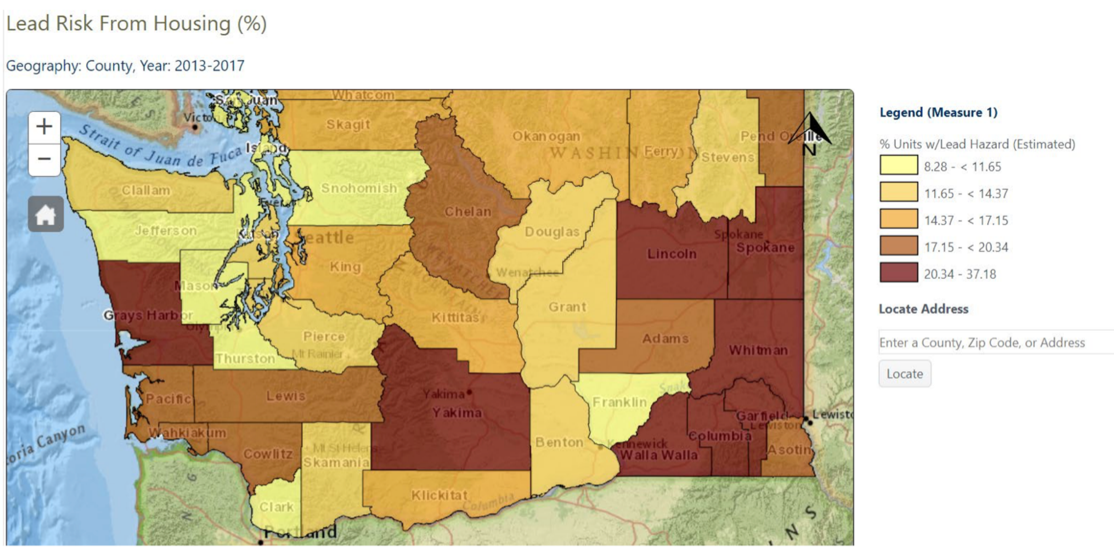
Housing Stock Characteristics

County-level housing data are drawn from the U.S. Census Bureau's American Community Survey (ACS), specifically Table B25034, which reports the number of housing units by decade of construction. This dataset allows us to identify the prevalence of older housing stock, which is highly correlated with lead-based paint risk.

For the purposes of this study, we define “old housing” as residential units constructed prior to 1989, consistent with the timeline of federal lead paint phase-out policies. Using this definition, we construct the variable *Old_House* as the proportion of total housing units in each county that were built before 1989. To reduce sampling variability and ensure comparability across counties, we utilize five-year ACS estimates spanning 2010–2023.

Figure 3 presents a spatial distribution of housing-based lead risk across Washington counties, as estimated by the Washington State Department of Health. The map reflects the most recent available data (2013–2017) and classifies counties by the percentage of housing units estimated to contain lead hazards. Darker shading indicates counties with higher housing-related lead risk, highlighting regional disparities in environmental vulnerability.

Figure 3. *Washington Tracking Network Map: Lead Risk from Housing*



Demographic and Socioeconomic Indicators

Demographic indicators—including total population, population density, and disaggregated counts by race and sex—are obtained from the Washington State Office of Financial Management (OFM). These data are available annually at the county level from 2000 to 2024, providing consistent coverage across all jurisdictions. Economic variables, including median household income, the number of individuals in poverty, and the poverty rate, are drawn from the U.S. Census Bureau’s Small Area Income and Poverty Estimates (SAIPE) program. SAIPE provides annual county-level estimates from 1995 to 2023, ensuring comparability over time and across counties.

Unemployment Data

County-level unemployment rates are obtained from the U.S. Bureau of Labor Statistics (BLS), which reports annual estimates from 2000 to 2024. These data serve as controls for local labor market dynamics that may influence both housing quality and public health outcomes, including lead exposure risk.

Variable Names

Table 3 lists all variables used in the analysis along with their definitions and data sources. These include the primary outcome variable, blood lead level (BLL) exceedance rates, and the estimated conditional average treatment effect (CATE) generated using the Causal Forest model. The dataset also incorporates demographic, socioeconomic, and housing-related controls, such as median household income, poverty rate, housing age, and unemployment rate. GRANT_AMOUNT captures the total HUD lead hazard control grant funding received by each county-year, and Grant_received is a binary indicator of program participation.

Table 4 presents descriptive statistics for the key variables. The average BLL exceedance rate among children under 72 months is relatively low (mean = 4.32%), but variation is substantial, with some counties experiencing rates exceeding 40%. Approximately 65% of the housing stock, on average, is classified as "old" (built prior to 1989), which aligns with known risk factors for lead-based paint exposure. The poverty rate averages around 13.75%, and population density ranges widely, from fewer than 5 to over 1,100 residents per square mile.

Demographic composition is similarly heterogeneous. On average, each county has about 88,946 female residents and 66,494 children under the age of six. The racial distribution is predominantly White, though several counties exhibit sizable Asian, Black, and Native populations. Grant funding is highly unevenly distributed: while the mean annual GRANT_AMOUNT is approximately $191,559, the standard deviation is over $1.7 million, with some counties receiving over $41 million and others receiving none.

This multidimensional heterogeneity provides strong justification for examining the conditional effects of lead hazard mitigation policies across diverse local contexts.

TABLE 3. Data and Name of Variables

Name of Variable	Data
Estimated_CATE_CF	Estimated effectiveness of grant by Causal Forest
BLL_ABOVE_THRESHOLD	Rate of children resulted BLL exceeding 5 µg/dL.
NUM_CHILDREN_UNDER_72MONTHS	Number of children under 72 months by county
Old_House	Percentage of house built prior to 1989 by county
MEDIAN_HOUSEHOLD_INCOME	Median household income by county
GRANT_AMOUNT	Amount of grant received from HUD by county
Grant_received	Dummy variable indicating the grant receiving
POVERTY_PERCENT	Percentage of poverty by county
UNEMP_RATE	Unemployment rate by county
BLACK TOTAL	Number of African American population in the county
ASIAN TOTAL	Number of Asian populations in the county
WHITE TOTAL	Number of White populations in the county
MALE	Number of Male populations in the county
POPULATIONS	Number of total populations in the county
YEAR	Year from 2000 to 2023
COUNTY_ID	Numerical value for each county by alphabetical order
COUNTY	County names

TABLE 4. Summary Statistics

	Obs.	Mean	Std. Dev.	Min	Max
BLL_ABOVE_THRESHOLD	446.0	4.32	3.95	0.3	40.0
NUM_CHILDREN_UNDER_72MONTHS	769.0	66493.74	127351.64	621.0	732600.0
GRANT_AMOUNT	781.0	191558.65	1790241.99	0.0	41330234.89
Old_House	585.0	0.65	0.09	0.42	0.9
POPDEN	976.0	134.45	221.96	3.17	1124.28
MEDIAN_HOUSEHOLD_INCOME	858.0	51948.69	14540.49	29066.0	120672.0
UNEMP_RATE	976.0	7.18	2.23	2.8	17.1
MALE	976.0	88419.83	176795.82	1118.01	1195327.0
FEMALE	976.0	88946.07	177586.23	1119.97	1182773.0
POVERTY_PERCENT	858.0	13.75	3.67	6.5	32.3
ASIAN TOTAL	976.0	14346.84	56584.01	11.0	519667.0
WHITE TOTAL	976.0	139982.56	249367.22	2121.0	1396927.0
BLACK TOTAL	976.0	6930.29	23003.31	0.0	171318.0

V. Methodology

The primary aim of this study is to estimate the heterogeneous effects of HUD-funded lead hazard control grants on childhood blood lead levels (BLLs) across counties in Washington State. Recognizing that policy effectiveness may vary significantly by local housing age, poverty rates, or demographic structure, we employ a machine learning–based causal inference framework to capture and quantify this heterogeneity.

Estimation Strategy: Causal Forest

To estimate **Conditional Average Treatment Effects (CATE)**, we employ the Causal Forest algorithm developed by Wager and Athey (2018). Causal Forests extend traditional random forests by allowing for causal effect estimation at the individual (here, county-year) level. The method recursively partitions the covariate space to maximize variation in treatment effects, rather than prediction accuracy.

Formally, for each observation i, the model estimates:

$$\hat{\tau}(X_{it}) = \mathbb{E}[\, Y_{it}(1) - Y_{it}(0) \mid X_{it} \,]$$

where $Y_i(1)$ and $Y_i(0)$ represent the potential outcomes under treatment and control, respectively. In this analysis, $Y_i(1)$ represent the percentage of children who exceed the blood lead level threshold with grant awarded in county i. $Y_i(0)$ represents the same measure but without grant awarded in county i. The covariate vector X_{it} includes a rich set of observed county-year characteristics that may influence both the likelihood of receiving a HUD grant and the level of lead exposure outcomes. Specifically, X_{it} comprises demographic indicators such as population density, racial composition, and sex ratios; socio-economic variables including median household income, poverty rate, and unemployment rate; housing stock features such as the proportion of homes built before 1989 (i.e., *Old_House*); and child-specific indicators including the number of children under 72 months and the number of lead tests conducted. By conditioning on these covariates, the Causal Forest model isolates variation in treatment effects attributable to structural and contextual differences across counties.

Although the Conditional Average Treatment Effect (CATE) is commonly expressed as $\hat{\tau}(X_i) = \mathbb{E}[\, Y_i(1) - Y_i(0) \mid X_i \,]$, this notation implicitly assumes that the covariate vector X_i captures all relevant characteristics of observational units. In the context of our panel dataset, where observations are indexed by both county and year, we refine this notation to X_{it} to emphasize that the covariates vary over time as well as across counties. Accordingly, the treatment effect we estimate is more accurately described as $\hat{\tau}(X_{it})$ reflecting the conditional impact of HUD grant receipt in county i during year t, given the observed county-year characteristics. This extension is especially important in our setting, where both treatment assignment (HUD grant receipt) and outcome variables (percentage of children's BLL test results exceeding the threshold) exhibit significant temporal variation. By conditioning on X_{it}, the Causal Forest model accounts for time-varying confounders and enables the estimation of dynamic, localized treatment effects across both space and time.

The model relies on the assumption of unconfoundedness conditional on observed covariates and employs sample splitting (i.e., honest trees) to mitigate overfitting. Since both potential outcomes— $Y_i(1)$ and $Y_i(0)$ —are never observed simultaneously for any individual, the Conditional Average Treatment Effect (CATE) is estimated using the Causal Forest method. This approach constructs an ensemble of randomized decision trees and aggregates localized treatment effect estimates across them, allowing for flexible and consistent estimation of treatment effect heterogeneity. The estimated CATE for unit i is computed as:

$$\hat{\tau}(X_i) = \frac{1}{B}\sum_{b=1}^{B} \hat{\tau}_b(X_i)$$

where B is the total number of trees in the forest (e.g., 500 in our setting) and $\hat{\tau}_b(X_i)$ denotes the estimated treatment effect in the b-th tree for unit i, or the local treatment effect estimated by the b-th tree. $\hat{\tau}(X_i)$ is the aggregated CATE estimate for covariate profile x. Each tree partitions the covariate space to isolate treatment and control units with similar characteristics, enabling data-driven estimation of heterogeneous effects.

We implement the algorithm using the *econml* package in Python, with random forests as the base learners for both the outcome and treatment models. Hyperparameters such as the number of trees, depth, and minimum leaf size are selected via cross-validation. To improve estimation efficiency, we apply feature selection using variance thresholds and standardize continuous covariates prior to model fitting.

Traditional Regression Model

To investigate the heterogeneous impact of HUD lead hazard control grants across Washington counties, we employed both Ordinary Least Squares (OLS) regression and Generalized Additive Models (GAM). These complementary approaches allow us to explore the structural and demographic factors that contribute to variations in policy effectiveness. This approach allows for flexible modeling of each explanatory variable's effect on CATE by fitting penalized spline functions rather than assuming strict linearity.

OLS with Fixed Effects

We estimate a two-way fixed effects panel regression model of the form:

$$CATE_{it} = \beta_0 + \sum_k \beta_k X_{kit} + \gamma_i + \delta_t + \varepsilon_{it}$$

where $CATE_{it}$ is the estimated Conditional Average Treatment Effect in county and year; X_{kit} denotes a vector of time-varying county-level covariates (including housing age, income, poverty, child population, unemployment, and racial composition); γ_i and δ_t are county and year fixed effects, respectively. The model was estimated using county and year dummy variables to absorb unobserved heterogeneity.

Generalized Additive Model (GAM)

To allow for nonlinear relationships between covariates and CATE, we also estimate a Generalized Additive Model:

$$E[\,Y_i \mid X_i\,] = \beta_0 + \sum_{j=1}^{p} f_j(X_{ij}) + \varepsilon_i$$

Where Y_i is the estimated CATE for observation i; X_{ij} represents the j-th covariate, $f_j(\cdot)$denotes a smooth spline function estimated for each predictor j and ε_i is the error term.

The GAM was implemented using the *pygam* library in Python. To avoid misinterpretation of smooth effects, we excluded categorical fixed effects for county and year.The GAM includes nine continuous predictors: the proportion of older housing, median household income, poverty rate, number of children under 72 months, unemployment rate, racial composition (Black, Asian, and White populations), and male population. Model performance was assessed using the coefficient of determination (R^2) and the Root Mean Squared Error (RMSE). Partial dependence plots were generated to visualize the marginal effects of each predictor on CATE.

Application to BLL Outcomes

In addition to estimating determinants of CATE, we replicated both the OLS and GAM specifications using the proportion of children testing above the blood lead level threshold (BLL_{it}) as the dependent variable. All independent variables and model structures remained unchanged, except for the substitution of in place of CATE on the left-hand side. This parallel modeling strategy enables us to compare predictors of program effectiveness and direct lead exposure outcomes.

VI. Results

We applied the Causal Forest DML algorithm to estimate the conditional average treatment effects (CATE) of HUD lead hazard control grants on childhood blood lead levels (BLL) across Washington State counties. The model was trained using a panel dataset that included demographic, economic, housing, and testing coverage variables for each county-year observation from 2000 to 2023.

The **estimated average treatment effect (ATE)** across all treated observations was approximately **–1.42 percentage point**, indicating that, on average, counties receiving HUD grants experienced a reduction in mean BLL relative to comparable counties without intervention. This effect size is consistent with the expected impact of environmental remediation policies and aligns with prior program evaluations (Clark et al., 2011). **Figure 4** shows the distribution of conditional average treatment effects of the grant programs for all counties. It shows highly left-skewed distribution with most values are in negative which evident that the HUD grant program was effective to reduce the overall BLL in counties. APPENDIX B illustrates the estimated treatment effect of grant program by county on the Washington State map. It provides more geographical view of the effect by each county.

***Figure 4**. CATE Distribution*

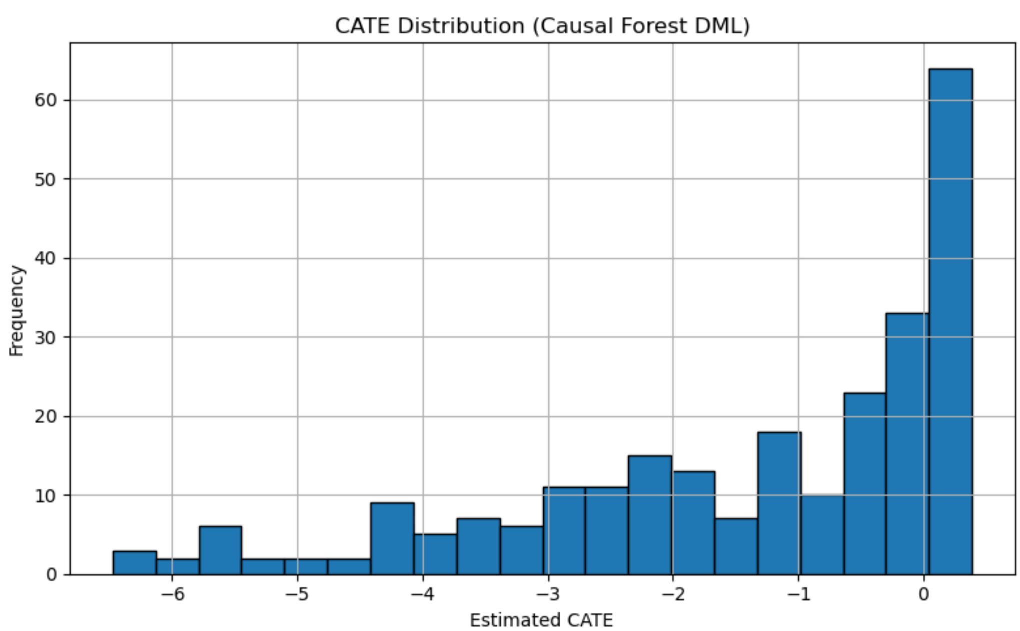


Figure 5 and 6 show top and bottom 10 counties with most and least effective grant program to the children's BLL test results. While the analysis show that the HUD program was effective to reduce the BLL in most counties, they show that each county experience different effect of the grant program. For instance, Garfield County (County number 12) shows the HUD program was the most effective with percentage of children tested above threshold decrease by 4.63 percentage point while King County shows only 0.18 percentage point decrease. **Table 5** includes the reduction in BLL of those top and bottom 3 counties. Garfield, Okanogan, and Columbia Counties showed the greatest declines in BLLs—each exceeding a 3.7% reduction. These counties are characterized by lower population density, older housing stock, and higher poverty rates, suggesting that lead abatement funding may be more effective in structurally vulnerable regions where baseline risks are higher and where relatively modest investments can yield substantial health gains. In contrast, King, Snohomish, and Clark Counties—urban and wealthier areas with newer housing infrastructure—showed minimal estimated reductions, each under 0.4%. These results underscore the importance of considering local housing and socioeconomic context when allocating lead hazard control resources. The heterogeneity in treatment effects supports the core hypothesis of this study: grant effectiveness is not uniform, and targeted interventions may produce more efficient and equitable outcomes than broad, undifferentiated funding strategies.

Figure 5. *Bottom 10 counties with least effective Grant*

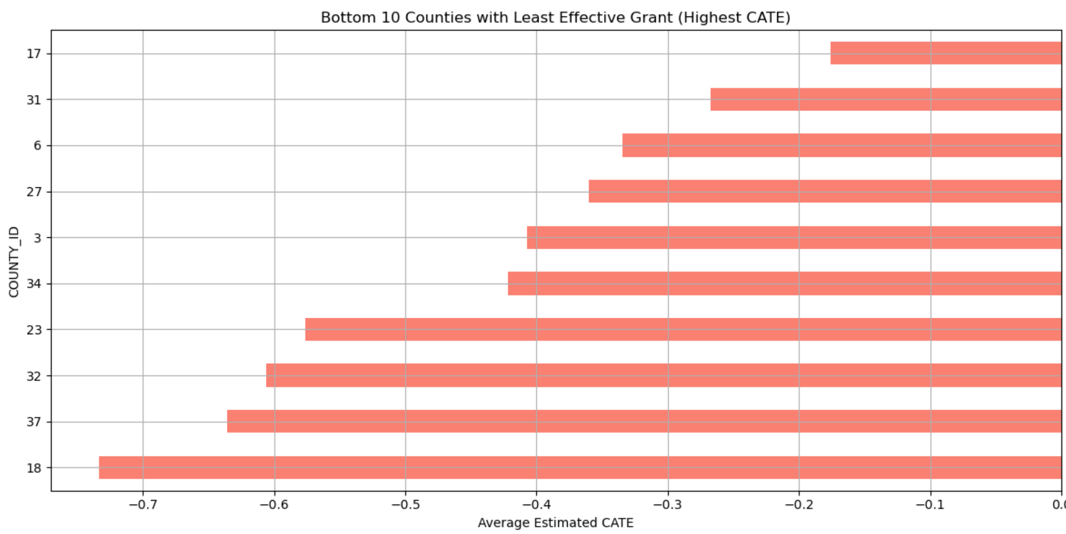


Figure 6. *Top 10 counties with most effective Grant*

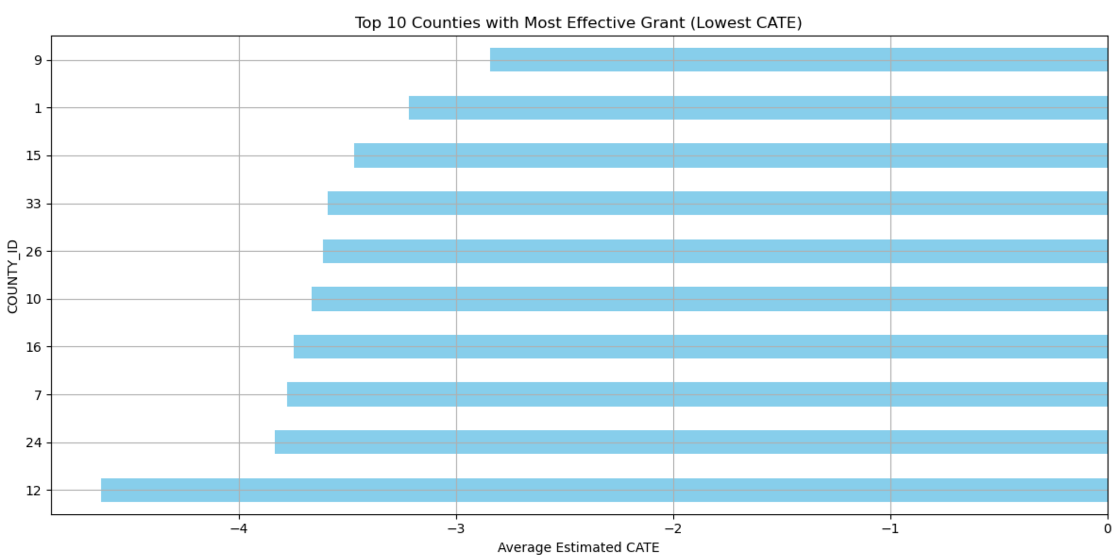


TABLE 5. Top and Bottom 3 counties with effect

Top 3 Counties (Largest Reductions)		**Bottom 3 Counties (Smallest Reductions)**	
Garfield County	-4.63%	King County	-0.18%
Okanogan County	-3.84%	Snohomish County	-0.27%
Columbia County	-3.78%	Clark County	-0.33%

NOTE: left columns indicate the name of county and right columns show the estimated CATE in percentage point.

Following **Table 6** shows the result of regression analysis of variables and the CATE to identify the reason behind the heterogeneity of the grant program. Each column shows different model with different type of variables. We ran three different OLS models with different variables.

TABLE 6: OLS Regression

Independent Variable: Estimated_CATE	**Model (1)**	**Model (2)**	**Model (3)**
Constant	-8.307*** (1.249)	-11.376*** (1.363)	-575.727*** (45.731)
Old_House	-3.311*** (1.106)	-4.022*** (1.034)	-2.124** (0.839)
MEDIAN_HOUSEHOLD_INCOME	0.0001*** (1.33e-05)	0.0002*** (1.62e-05)	5.48e-05*** (1.59e-05)
POVERTY_PERCENT	0.131*** (0.032)	0.215*** (0.033)	0.060* (0.031)
NUM_CHILDREN_UNDER_72MONTHS	-1.60e-06* (8.19e-07)	-7.19e-07 (1.15e-06)	1.47e-07 (9e-07)
POPDEN	-0.0004 (0.001)	-0.0013** (0.001)	0.0003 (0.0004)
MALE		-1.28e-05 (1.51e-05)	-3.85e-05*** (1.22e-05)
BLACK_TOTAL		6.29e-07 (1.82e-05)	9.11e-06 (1.46e-05)
ASIAN_TOTAL		-1.07e-05 (8.43e-06)	9.08e-06 (6.98e-06)
WHITE_TOTAL		1.15e-05 (7.78e-06)	2.54e-05*** (6.22e-06)
COUNTY_ID			0.0065 (0.006)
YEAR			0.284*** (0.023)
R-squared	0.444	0.557	0.732

*NOTE: Robust standard errors in parentheses. Significance levels: * p<0.1, ** p<0.05, *** p<0.01. These three models were not run with fixed effects due to model overfitting problem.*

In the first model, the coefficient of percentage of old-house, median household income and poverty rate are statistically significant. However, only percentage of old-house and poverty percent show meaningful value. It estimates that 1% increase in rate of old house reduces about 3% of children who tested above the BLL threshold level when they receive the grant. Although the relationship between percentage of poverty and CATE are not strong, as it indicates 0.131, first model shows that 1% increase in poverty rate decreases the effect of policy. Model (2) and Model (3) show similar result. Only old house shows significant impact to effectiveness of the HUD grant program. Three models commonly evident that the effect of policy is stronger in the locations where there is more old-house rate.

In addition to linear OLS regression analysis, we added non-linear regression analysis by using The Generalized Additive Model (GAM) method. The GAM reveals strong non-linear associations between estimated treatment effects (CATE) and several covariates. Notably, the smooth terms for **median household income**, **poverty rate**, and **racial composition (Asian and White population)** all show high statistical significance ($p < 0.001$). The effective degrees of freedom (EDF) for these terms suggest complex, non-linear relationships. The model achieves a **Pseudo R^2 of 0.85**, indicating excellent fit, with relatively low generalized cross-validation error (GCV = 0.93), further confirming model robustness. **Table 7** shows the summary of GAM model with statistical result. It shows that old house, median household income, poverty rate, unemployment rate, level of Asian and white population are statistically significant.

Figure 7 presents the partial dependence plots (PDPs) for each predictor variable from the Generalized Additive Model (GAM). These plots visualize the estimated smooth effects of each covariate on the Conditional Average Treatment Effect (CATE), while holding all other variables constant. The blue line represents the estimated marginal effect of each variable on the policy outcome, and the red dashed lines denote the 95% confidence intervals. Nonlinear patterns, such as the inverted-U shape for median household income and the saturation effects in unemployment and poverty, reveal heterogeneous treatment effects across different socioeconomic and demographic profiles.

Notably, the effect of *Old_House* shows a declining trend, suggesting stronger policy impact (lower CATE) in areas with a higher proportion of older housing. Also, the plot in MEDIAN_HOUSEHOLD_INCOME show that the grant is more effective in counties where have relatively lower median household income level. As the plot shows, the marginal effect of the grant program is not severe at the certain level of median income level, in this case, about $70,000.

In contrast, the partial dependence plot for *POVERTY_PERCENT* reveals a clear **positive and nonlinear relationship** between local poverty levels and the estimated CATE. As poverty rates increase, the estimated treatment effect (CATE) becomes progressively **less negative**, suggesting that the **marginal effectiveness of the policy diminishes** in higher-poverty areas. Notably, in counties with **low to moderate poverty rates (below ~20%)**, HUD grants appear to have **strong negative CATE values**, indicating **substantial reductions in lead exposure**. However, beyond the **25–30% poverty threshold**, the policy's marginal benefit flattens or even weakens, possibly due to **diminishing returns, implementation challenges**, or unmeasured compounding risk factors in high-poverty environments. This result suggests that while HUD grants are **generally more effective in poorer areas**, the **most disadvantaged counties** may require **complementary interventions** (e.g., health outreach, education, or enforcement mechanisms) to fully realize policy impact.

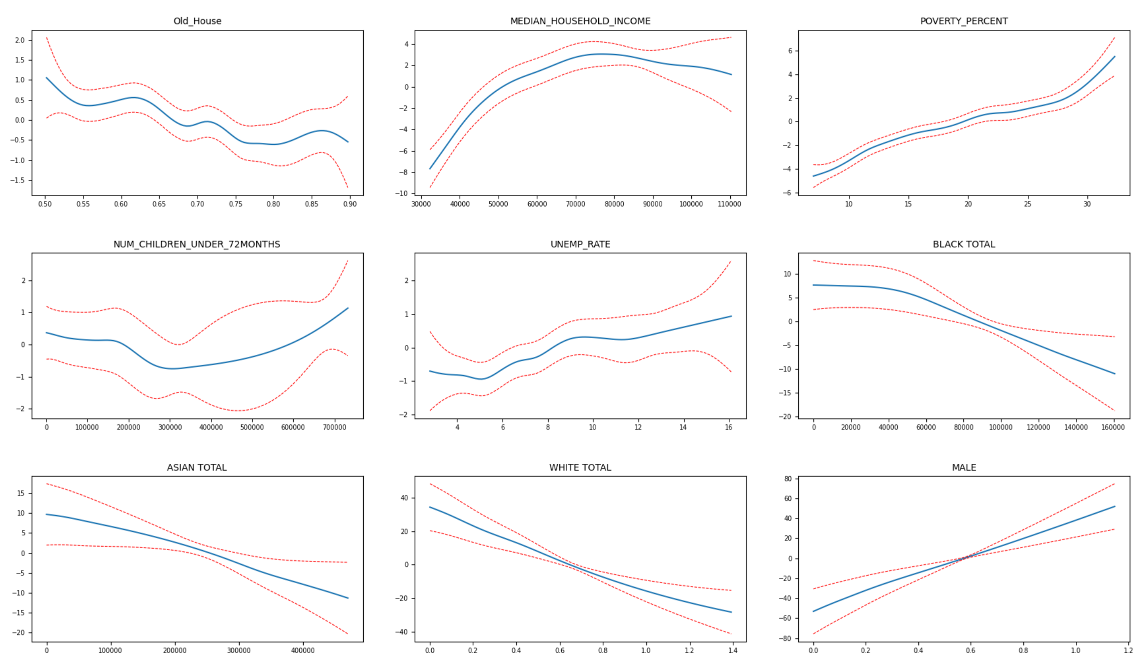
The partial dependence plot for *UNEMP_RATE* indicates a **positively sloped nonlinear relationship** between local unemployment levels and the estimated treatment effect (CATE). Specifically, as unemployment rises, the estimated CATE becomes **less negative**, suggesting a **decrease in policy effectiveness** in higher-unemployment areas. In regions with **low to moderate unemployment rates (3–8%)**, the grant's impact on reducing lead exposure appears more pronounced, as indicated by lower (more negative) CATE values. However, as unemployment exceeds **10–12%**, the slope steepens and the effect flattens near zero, implying **diminishing returns** or **structural barriers** that may hinder policy implementation in economically depressed regions. These results imply that while unemployment is often associated with vulnerability, **excessively high unemployment may coincide with systemic challenges** (e.g., housing instability, lack of institutional support) that weaken the impact of HUD interventions.

Table 7. Generalized Additive Model (GAM) Summary

Term	EDF	p-value	Significance
s(Old_House)	12.9	0.000279	***
s(MEDIAN_HOUSEHOLD_INCOME)	11.5	<0.0001	***
s(POVERTY_PERCENT)	8.7	<0.0001	***
s(NUM_CHILDREN)	9.3	0.029	*
s(UNEMP_RATE)	6.4	0.000948	***
s(MALE)	2.9	0.176	
s(BLACK TOTAL)	0.7	0.096	.
s(ASIAN TOTAL)	0.5	0.000173	***
s(WHITE TOTAL)	0.3	0.000172	***
Intercept	-	0.387	
Pseudo R^2	0.850		
AIC	632.2		
GCV	0.928		

NOTE: EDF = Effective Degrees of Freedom. Higher EDF indicates greater flexibility and non-linearity in the fitted smooth term. GCV = Generalized Cross-Validation score, a measure of model predictive performance; lower is better. P-values are approximate and may be underestimated due to smoothing parameter selection.

***Figure** 7. Partial Effects of Policy by Variable (GAM Model results)*



NOTE: Partial dependence plots capture average marginal effects and do not account for higher-order interactions unless explicitly modeled. Interpretations should consider potential omitted variable bias and heterogeneity.

Table 8 presents results from two Ordinary Least Squares (OLS) regressions estimating the associations between socioeconomic, demographic, and housing characteristics and the rate of children testing blood lead levels (BLL) exceeding the threshold. Both models include county and year fixed effects to control for time-invariant regional characteristics and unobserved temporal shocks. Model (1) includes a core set of explanatory variables related to housing age, income, poverty, unemployment, and child population. Model (2) extends this baseline specification by adding demographic composition variables, including racial and gender distributions. Robust standard errors are reported in parentheses, and asterisks denote statistical significance at the 10%, 5%, and 1% levels.

Table 8. OLS Regression Results with Fixed Effects (Time and County)

VARIABLES	Model (1)	Model (2)
Constant	11.294*** (3.186)	18.164*** (3.605)
Old_House	6.256** (2.479)	7.053*** (2.474)
MEDIAN_HOUSEHOLD_INCOME	-0.0002*** (3.08e-05)	-0.0002*** (4.01e-05)
POVERTY_PERCENT	-0.109 (0.070)	-0.281*** (0.079)
NUM_CHILDREN_UNDER_72MONTHS	2.46e-06 (1.52e-06)	-3.63e-06 (2.79e-06)
UNEMP_RATE	-0.353*** (0.110)	-0.336*** (0.107)
BLACK TOTAL		-4.64e-05 (4.36e-05)
ASIAN TOTAL		-4.39e-06 (2.02e-05)
WHITE TOTAL		-3.30e-05* (1.86e-05)
MALE		5.91e-05 (3.62e-05)
R-squared	0.598	0.613

*NOTE: Robust standard errors in parentheses. Significance levels: * p<0.1, ** p<0.05, *** p<0.01. Both models include fixed effects for COUNTY_ID and YEAR. Both models include time and entity fixed effects.*

In both specifications, the proportion of **older housing (Old_House)** is a statistically significant and robust predictor of elevated BLL, with estimated coefficients of **6.256** in Model (1) and **7.053** in Model (2). This finding is consistent with the known risk of lead-based paint in pre-1980 housing stock. Likewise, **median household income** is negatively associated with BLL in both models ($p < 0.01$), indicating that children in lower-income households are at significantly higher risk of lead exposure.

The effect of **poverty rate** differs across models. While not significant in Model (1), it becomes strongly negative and significant in Model (2) (coef = -0.281, $p < 0.01$). This reversal may reflect interaction effects with added demographic variables or multicollinearity. Interestingly, **unemployment rate** is also negatively associated with BLL, contrary to conventional expectations, suggesting possible confounding factors such as migration patterns, vacancy rates, or differential access to healthcare and testing in high-unemployment regions.

Among demographic controls introduced in Model (2), the proportion of **White residents** exhibits a marginally significant negative association with BLL ($p < 0.1$), while other racial and gender indicators are not statistically significant. These results underscore that **housing conditions and income levels** remain the most critical predictors of lead exposure risk, while demographic composition has a more limited or context-dependent effect.

Overall, the findings reinforce the importance of prioritizing **older, low-income neighborhoods** in lead abatement and grant allocation policies. Moreover, the unexpected signs on variables such as poverty and unemployment suggest that **structural disadvantages may not always align linearly with environmental health risks**, pointing to the need for more granular or interaction-based policy analysis.

VII. Interpretation:

The empirical findings of this study reveal substantial heterogeneity in the effectiveness of HUD's lead hazard mitigation grants across counties in Washington State. Most notably, the presence of older housing stock consistently emerged as the most influential determinant of policy impact across all modeling approaches. Both the Causal Forest algorithm and traditional regression analyses demonstrate that grants were more effective in counties with a higher proportion of pre-1989 homes—structures more likely to contain lead-based paint. This finding aligns with the epidemiological literature linking deteriorated lead paint in aging homes to elevated blood lead levels in children.

OLS and GAM models alike underscore this pattern. The partial dependence plot for the variable *Old_House* reveals a downward-sloping marginal effect, suggesting that HUD interventions yield greater reductions in BLL in counties with more lead-risk housing. The significance and direction of this relationship were robust across specifications, suggesting a well-targeted allocation strategy in terms of physical infrastructure vulnerability.

The analysis also surfaces nuanced relationships between grant effectiveness and socioeconomic conditions. Median household income showed an inverted-U pattern in the GAM model, with HUD grants being most effective in counties with lower to moderate income levels (particularly below $70,000). This suggests that extremely low-income areas may lack complementary conditions—such as institutional capacity or stable tenancy—that enable effective implementation, while high-income areas may already have fewer risk exposures, diminishing marginal returns from federal funding.

Poverty rate and unemployment exhibited diminishing marginal returns as their levels increased. In counties with moderate poverty (10–20%), HUD grants were highly effective. However, at the upper tail of the poverty distribution (above 25–30%), the effectiveness plateaued or declined. This may reflect structural barriers in high-poverty counties—such as lower enforcement capacity, weaker landlord compliance, or higher baseline risk factors—that reduce the efficacy of financial interventions alone. Similarly, the positive slope observed in the unemployment-rate PDP suggests that high-unemployment counties benefit less from HUD funding, likely due to compounding vulnerabilities that blunt the effectiveness of isolated policy efforts.

The racial composition variables also displayed statistically significant, albeit modest, nonlinear associations with CATE. Counties with moderate levels of Asian and White populations experienced larger grant effects, though the mechanisms behind this remain speculative and may reflect correlations with unobserved factors such as neighborhood segregation, rental market dynamics, or historical disparities in code enforcement.

In parallel models using BLL exceedance rate as the outcome, the same key drivers—old housing, income, poverty, and unemployment—retained significance, reinforcing their importance not just in determining where policy is effective, but also where need is greatest. The robustness of these findings across models and outcomes supports a broader implication: **place-based lead hazard interventions are most successful when they align with structural vulnerability, but they must also be supported by implementation capacity and local engagement to maximize impact.**

Together, these results highlight the importance of refining HUD's targeting strategy to account not only for housing risk but also for local socioeconomic conditions. Grants alone are unlikely to be uniformly effective across all contexts; rather, their success depends on tailoring interventions to local realities. In high-risk, high-poverty areas, complementary investments in outreach, enforcement, or tenant protections may be necessary to ensure equitable reductions in childhood lead exposure.

VIII. Limitations and Directions for Future Research:

While this study provides novel evidence on the heterogeneous effectiveness of HUD-funded lead hazard interventions in Washington State, several limitations warrant discussion.

First, the blood lead level (BLL) data are incomplete across both time and geography. Inconsistent reporting of child BLL screening results—particularly in earlier years and in less populous counties—reduces the statistical power and representativeness of our estimates. Future research would benefit from access to individual-level microdata or the application of small-area estimation techniques, including Bayesian spatial smoothing or imputation methods, to enhance coverage and precision.

Second, information on housing conditions—especially the age of residential structures—is temporally coarse. The ACS Year Structure Built (B25034) variable is available only in decade-level bins and limited to post-2010 releases. This restricts our ability to track changes in housing risk over time and may attenuate the observed treatment heterogeneity associated with aging housing stock.

Third, data on HUD grant allocations lack intra-county resolution. Although award amounts and recipient organizations are publicly available, the spatial distribution of expenditures within counties is not. Without tract-level allocation records, we cannot assess whether resources were evenly distributed or concentrated in high-risk neighborhoods, limiting our capacity to estimate within-county treatment variation and potentially biasing effect estimates toward zero.

Fourth, several key control variables—such as county-level poverty rates and median household income—are missing for the years 2010 and 2011. These gaps overlap with critical post-recession years, introducing potential bias in covariate adjustment.

Fifth, the analysis is restricted to Washington State, which may limit the external validity of the findings. The nature of lead exposure risk, as well as the targeting and implementation of HUD-funded programs, likely varies across states. Future studies should extend this framework to multi-state or national panel datasets to evaluate generalizability and cross-state policy effectiveness.

Finally, this study does not account for spatial spillovers or displacement effects. HUD interventions may affect not only the treated areas but also surrounding neighborhoods through migration, housing turnover, or changes in investment dynamics. Incorporating geocoded grant-level data and applying spatial econometric models could yield valuable insights into the broader regional impact of policy interventions.

IX. Conclusion:

This study evaluated the heterogeneous effects of HUD-administered lead hazard reduction grants across the 39 counties in Washington State using both traditional econometric and modern machine learning approaches. By applying Causal Forest models to a panel dataset spanning 2000 to 2023, we uncovered significant spatial variability in policy effectiveness—some counties saw meaningful reductions in childhood blood lead levels, while others exhibited negligible or even adverse outcomes.

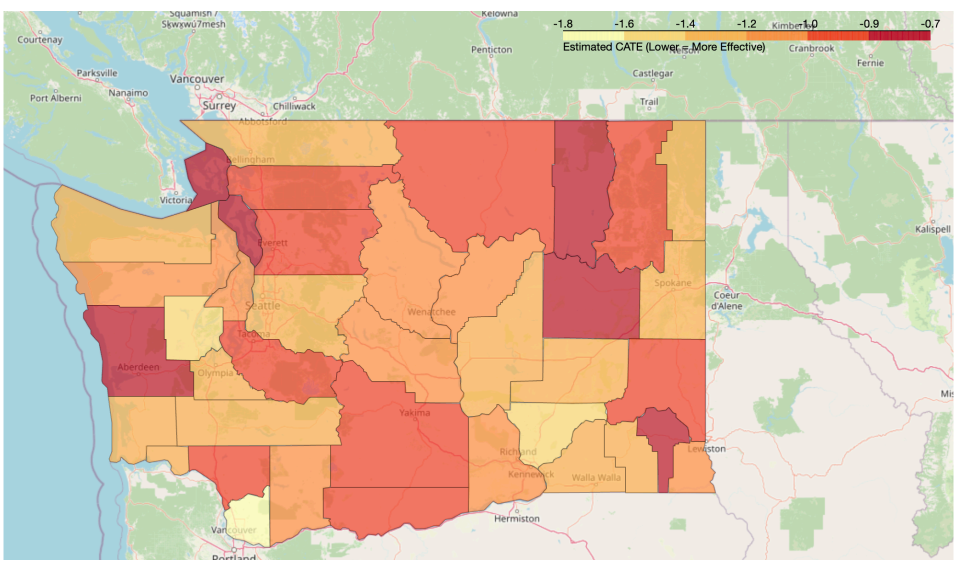
Our analysis demonstrates that policy impact is not uniform, but rather mediated by structural conditions such as housing age, poverty rates, and demographic composition. Notably, rural counties like Garfield, Columbia, and Okanogan experienced the most substantial improvements, suggesting that targeting grants toward high-risk, underserved areas may yield more pronounced health benefits. In contrast, more urbanized counties such as King and Snohomish—despite receiving substantial funding—showed minimal changes, possibly due to implementation barriers, existing infrastructure complexity, or diminishing marginal returns.

Methodologically, the integration of causal machine learning techniques enabled us to move beyond average treatment effects, allowing for a nuanced understanding of where and for whom public health interventions are most effective. Comparing OLS and GAM models further clarified which contextual variables drive the heterogeneity in outcomes, reinforcing the need for adaptive, data-driven grant allocation mechanisms.

These findings underscore the importance of spatially informed environmental policy. As lead exposure risks persist despite decades of regulatory progress, targeted remediation efforts—guided by rigorous impact evaluation and localized data—can serve as powerful tools to close public health gaps. Policymakers should not only expand funding but also refine allocation strategies based on empirical evidence of need and responsiveness. This approach will help ensure that federal lead abatement initiatives fulfill their promise of protecting all children—regardless of where they live.

APPENDIX

APPENDIX A. Washington State map with County Names



APPENDIX B. Estimated treatment effect across Washington state and couties

Pseudo-code: Estimating Heterogeneous Treatment Effects using Causal Forest

```
Input:
    Dataset D = {(X₁, T₁, Y₁), ..., (Xₙ, Tₙ, Yₙ)}
    where:
        Yᵢ = BLL_ABOVE_THRESHOLD (lead exposure level)
        Tᵢ = grant_received (1 if HUD grant received, 0 otherwise)
        Xᵢ = covariates (e.g., income, poverty, old housing, etc.)
Procedure:
1. Preprocess Data:
    - Remove rows with missing values
    - Normalize or encode variables as needed

2. Initialize Random Forest models:
    - model_y: fits Y ~ X using a regression forest (e.g., RandomForestRegressor)
    - model_t: fits T ~ X using a classifier (e.g., RandomForestClassifier)

3. Train Causal Forest:
    - For b = 1 to B trees:
        a. Randomly sample observations and covariates
        b. Split data to maximize treatment effect heterogeneity
        c. In each leaf node, estimate local CATE:
            τ_b(x) = E[Y | T=1, X ∈ leaf] - E[Y | T=0, X ∈ leaf]

4. Aggregate across trees:
    - For each observation i:
        τ̂(xᵢ) = (1/B) ∑_{b=1}^{B} τ_b(xᵢ)

5. (Optional) Summarize results:
    - Compute ATE = E[τ̂(x)]
    - Visualize distribution of τ̂(x)
    - Analyze HTE across groups (e.g., poverty level, housing age)

Output:
    - Individual-level CATE estimates: τ̂(x₁), ..., τ̂(xₙ)
    - Group summaries of policy effectiveness
```

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